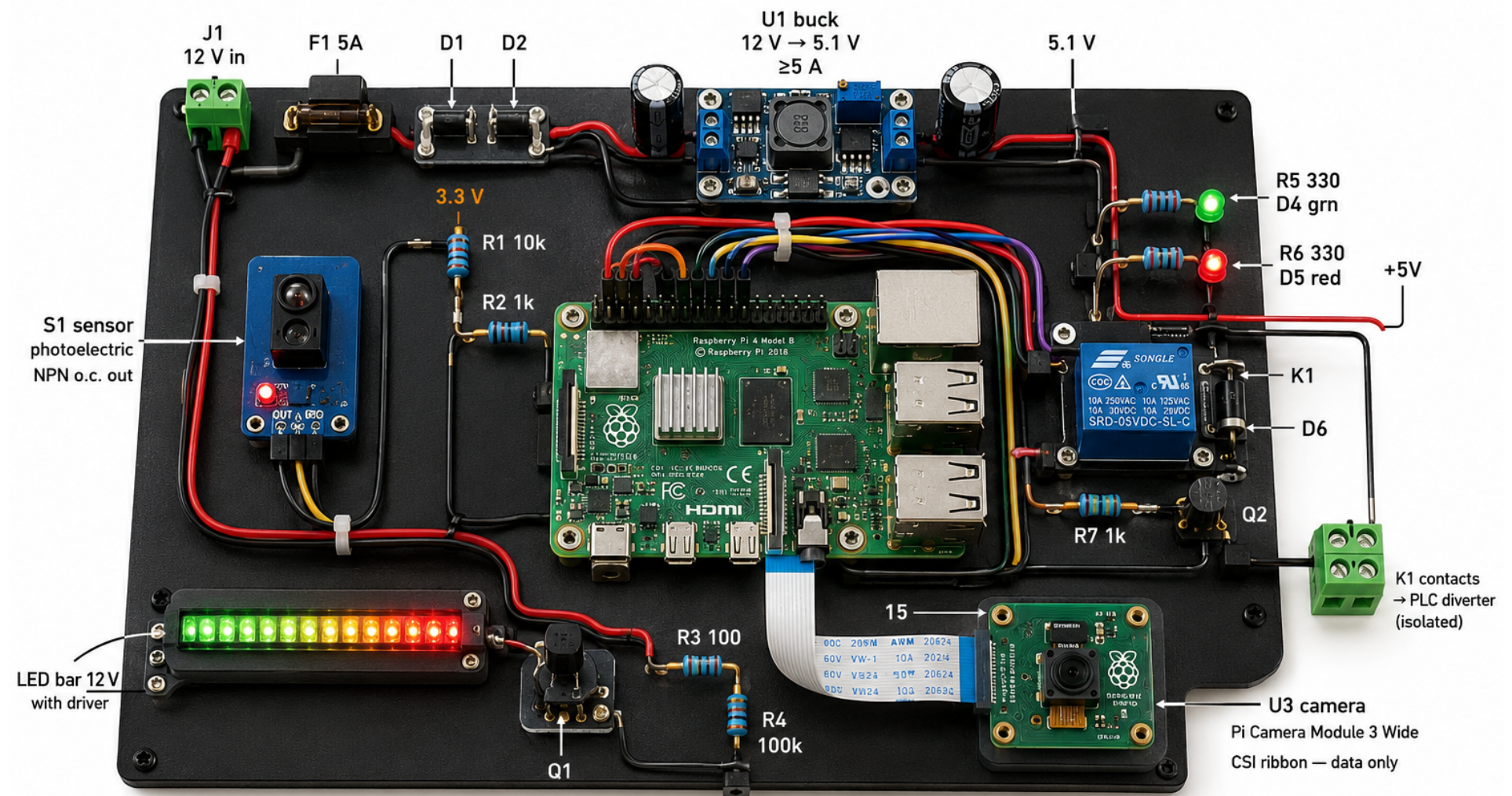


Real Object Drawing

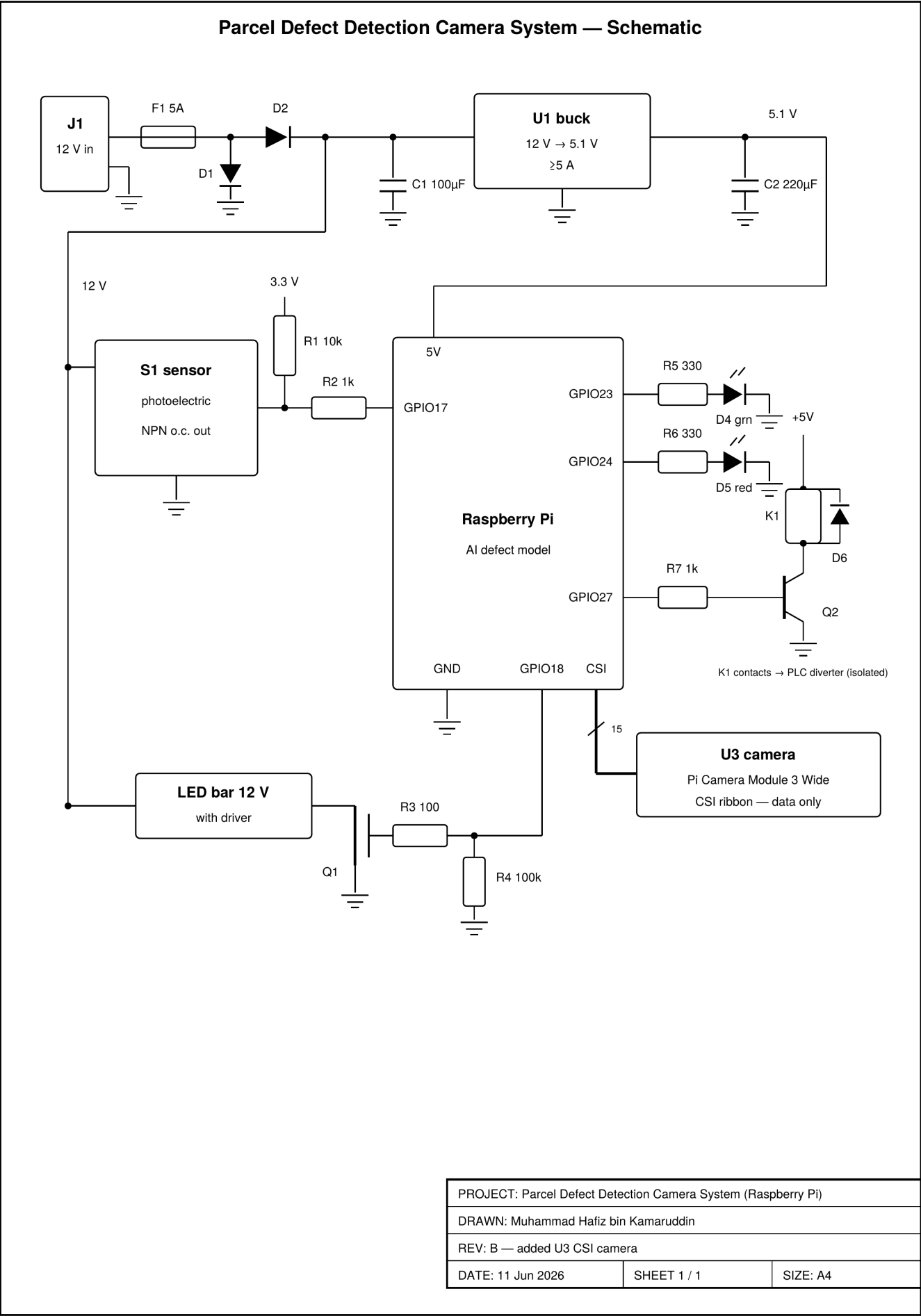
Physical layout of Raspberry Pi parcel-defect detection camera system



PROJECT: Parcel Defect Detection Camera System (Raspberry Pi)		
DRAWN: Muhammad Hafiz bin Kamaruddin		
REV: B — added U3 CSI camera		
DATE: 11 Jun 2026	SHEET 1 / 1	SIZE: A4

Schematic Diagram

Electrical schematic showing power, sensor, camera, LED and relay connections



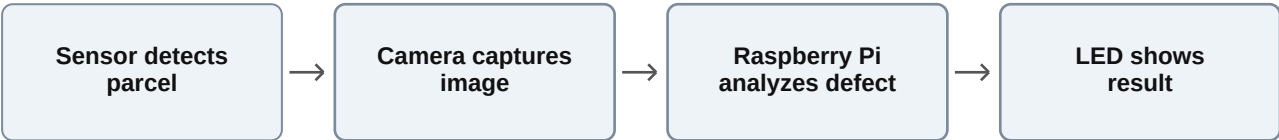
Main Component Functions

Short description of the sensor, camera, Raspberry Pi and LED

This page summarizes the role of the four main components used in the parcel defect detection system.

Component	Function
Sensor	Detects when a parcel arrives at the inspection area. It sends a trigger signal to the Raspberry Pi so the checking process can start.
Camera	Captures the parcel image. The image is used by the AI defect detection model to check whether the parcel is normal or defective.
Raspberry Pi	Acts as the main controller. It receives the sensor signal, captures and processes the camera image, runs the AI defect model, and decides the output result.
LED	Provides lighting so the camera can capture a clear image. The indicator LEDs can also show system status, such as pass/normal or defect/fail.

Simple system flow



In operation: the sensor detects the parcel, the camera captures the image, the Raspberry Pi runs the AI analysis, and the LED/indicator provides the visual result.